

MRI Super-Resolution for Low-Field Brain Imaging

Tumor-aware super-resolution: why image-quality metrics are the wrong target for safe enhancement

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Summary

Low-cost, portable MRI scanners produce lower-quality, blurry images. Machine learning models that sharpen these images can inadvertently remove a small tumor or introduce one that is not real. We develop a sharpening method trained to preserve tumors, and we measure how often each method hides or invents one, so that a low-quality scan can be made clearer without losing the detail a clinician needs to see.

Abstract

Portable low-field MRI could widen access to brain imaging where scanners are scarce, but its scans are noisy and low-resolution, and they look different from the high-field scans that analysis models are trained on (they are out-of-distribution), which hurts automated analysis. Deep learning super-resolution sharpens these scans to bridge that gap, but such models are usually trained and judged only by image-quality scores (PSNR, SSIM). Those scores barely change if the model smooths away a small tumor (pseudo-normalization) or invents one that is not there (hallucination), two failure modes expected from the perception-distortion tradeoff (Blau and Michaeli, 2018) and documented for deep medical imaging (Antun et al., 2020; Cohen et al., 2018; Bhadra et al., 2021). We simulate low-field scans by discarding high-frequency detail in the frequency domain (k-space) and adding realistic MRI noise, and train two super-resolution models: a standard image-quality-optimized one and a tumor-aware one that weights the tumor region. With image quality held equal, we measure how often each model erases a real tumor or invents a false one, broken down by tumor size, using a downstream tumor-segmentation model, and we produce an uncertainty map (via Monte Carlo dropout) that flags where the output is unreliable. We create the training pairs by degrading good scans ourselves, so no special paired or African dataset is required.

Contributions

- We define a physics-informed forward model that converts a high-resolution scan into a simulated low-field one by discarding the outer frequency-domain (k-space) detail and adding Rician noise. The degradation runs on the fly, so every low-resolution input keeps an exact high-resolution reference and no paired low-field or African dataset is required. Resolution loss and noise are reproduced; field-strength contrast change is not, which bounds the claims to simulated degradation.
 - We show that distortion-optimal super-resolution is unsafe for pathology: at matched image quality it suppresses small lesions, so PSNR and SSIM are inadequate targets for safety-critical enhancement.
 - We propose a tumor-aware (lesion-weighted) super-resolution objective and isolate its effect by comparing it against a distortion-optimal baseline at matched PSNR and SSIM.
 - We quantify two safety rates for the super-resolution-then-segmentation pipeline, a False Negative Erasure Rate and a False Positive Detection Rate, both stratified by lesion size.
 - We pair the model with Monte Carlo dropout uncertainty, test whether it predicts erasure, and verify CPU-only deployability.
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Definitions

- **MRI (magnetic resonance imaging):** imaging of soft tissue using magnetic fields, with no ionizing radiation.
- **Field strength (tesla, T):** the strength of the scanner’s static magnetic field. Higher field yields more signal and finer detail.
- **Low-field MRI:** scanners below roughly 0.5 T. They cost less and can be moved, and they produce noisier, lower-resolution images. Clinical high-field systems run at 1.5 T to 3 T.
- **k-space:** the frequency-domain representation of an MR image. The center carries low frequencies (coarse structure) and the edges carry high frequencies (fine detail). A scanner samples k-space and the image is its inverse Fourier transform.
- **Rician noise:** the distribution that MRI magnitude images follow. It arises when independent Gaussian noise on the real and imaginary channels is converted to a magnitude, and it departs from Gaussian behavior at low signal-to-noise ratio (Gudbjartsson and Patz, 1995).
- **Out-of-distribution:** an input unlike the data a model was trained on. Low-field scans are out-of-distribution for models trained on high-field scans.
- **BraTS (Brain Tumor Segmentation challenge):** a public multi-modal brain MRI dataset with expert tumor masks (Menze et al., 2015). Its three nested regions are Whole Tumor (all abnormal tissue including edema), Tumor Core (enhancing plus necrotic core), and Enhancing Tumor (the active enhancing part, which is small or absent in low-grade cases).
- **Super-resolution (SR):** recovering a higher-resolution image from a low-resolution one.
- **Forward model (degradation):** the operator that maps a high-resolution scan to a simulated low-field one. Ours truncates k-space and then adds Rician noise.
- **Truncation factor:** how severely k-space is cropped. Factor 4 keeps the central quarter of each frequency axis and discards the rest.
- **U-Net:** a convolutional encoder-decoder network with skip connections joining matching levels (Ronneberger et al., 2015).
- **Residual prediction:** the network outputs a correction that is added to its input rather than outputting the image directly.
- **Distortion-optimal:** trained only to minimize pixel error. This is our baseline objective.
- **Tumor-aware (lesion-weighted):** pixel error inside the tumor mask is multiplied by a factor, so an error on the lesion costs more than the same error in healthy tissue.
- **Monte Carlo dropout:** keeping dropout active at inference and averaging several stochastic passes. The spread across passes approximates predictive uncertainty (Gal and Ghahramani, 2016).
- **PSNR (peak signal-to-noise ratio, dB):** pixel fidelity against the reference image. Higher is better.
- **SSIM (structural similarity):** perceptual similarity against the reference, in $[0, 1]$. Higher is better.
- **Dice:** overlap between a predicted mask and the true mask, in $[0, 1]$. Higher is better.
- **Pseudo-normalization:** enhancement that makes an abnormal scan look normal by smoothing pathology away.
- **False Negative Erasure Rate (FNER):** the fraction of real lesions the pipeline no longer detects after enhancement.
- **False Positive Detection Rate (FPDR):** the rate at which tumor-like structure is detected where none exists (hallucination).
- **Matched quality:** comparing the two objectives at equal PSNR and SSIM, so that any difference in safety is attributable to the objective and not to sharpness.
- **AUROC (area under the receiver operating characteristic curve):** used here to test whether the uncertainty map ranks erroneous pixels above correct ones. A value of 0.5 is chance.

Motivation and related work

MRI is scarce across sub-Saharan Africa: fewer than one scanner per million people, of which about 39% are obsolete low-field systems that produce low-resolution images, against up to 37 per million in high-income countries (Anazodo et al., 2024). Low-field scanners are cheaper and more portable, which makes them a realistic path to access; the catch is image quality. A study in *Nature* showed deep learning can enhance 0.055T brain MRI, trained on public high-field scans with the low-field characteristics simulated (Man et al., 2023), and the accompanying commentary warns that such enhancement can hallucinate structure or erase abnormalities (*pseudo-normalization*) and needs validation (Johnson and Lui, 2023).

Super-resolution is known to improve downstream tumor segmentation (Zhou et al., 2022), and uncertainty estimation via Monte Carlo dropout (Gal and Ghahramani, 2016) has been applied to safer neuroimage enhancement (Tanno et al., 2021). What is missing is the observation that the standard training objective is itself misaligned with safety: an image-quality loss barely penalizes erasing a small tumor, so optimizing it does not protect pathology. We show this, fix it with a tumor-aware objective evaluated at matched image quality, and back it with an uncertainty map and a CPU-feasibility check. The corrective finding, not a new architecture, is our contribution. We demonstrate on standard open data; an African dataset is an optional bonus, not a requirement.

Method

Degradation (physics-informed forward model). We generate low-resolution and high-resolution training pairs by degrading good scans, so the original image is always the answer key. Two steps mimic a low-field scan rather than a plain blur: (1) *resolution loss* by cropping the center of k-space (the frequency-domain image), keeping the low-frequency block and dropping the high-frequency edges, then inverting; (2) *realistic noise* by adding Rician noise, the model MRI magnitude images actually follow.

What k-space is, in plain terms. Every image can be described in two equivalent ways. The familiar way is a grid of pixels. The other way is as a sum of ripples: a smooth broad wave that carries the overall shape, plus progressively finer ripples that carry the edges and the small details. The Fourier transform converts between the two descriptions, and neither one loses information, so an image can be moved back and forth freely. That second description is what radiologists call k-space, and an MRI scanner is unusual in that it measures k-space directly. The scanner does not record pixels and then compute the ripples; it records the ripples and the pixels are computed afterwards. The coarse ripples sit at the center of k-space and the fine ones sit at the edges. A weaker scanner cannot collect the outer ripples in a reasonable scan time, so it comes home with the center and none of the rim, and the reconstructed image looks blurred. Our degradation step does exactly that on purpose: it converts a sharp scan into ripples, throws the outer ones away, and converts back. The result blurs the way a real scanner blurs rather than the way a photo-editing blur filter does, which matters because the two leave different traces for a network to learn from.

Two super-resolution models. We train two small U-Nets with dropout: a *quality-optimal* baseline with a standard pixel loss, and a *tumor-aware* variant whose loss up-weights the tumor region (optionally with a segmentation-consistency term). Keeping dropout active at inference and running several passes (Monte Carlo dropout) yields a spatial uncertainty map.

Downstream segmentation. We segment the tumor with a compact 2D U-Net we train on BraTS. Training our own 2D model avoids the dimensionality and four-modality mismatch of large pre-trained 3D models. Segmentation is reported on the three standard nested regions: Whole Tumor (all abnormal tissue including edema), Tumor Core (enhancing plus necrotic core), and Enhancing Tumor (the active enhancing part, small or absent in low-grade cases).

Model per stage. Every network is trained from scratch. No pretrained weights are used, and one shared U-Net implementation serves both tasks so that the objective is the only variable under test. We deliberately avoid an adversarial (GAN) objective: adversarial losses reward output that merely looks like plausible tissue, which is the documented mechanism behind hallucinated structure (Co-

hen et al., 2018; Bhadra et al., 2021), so a GAN would manufacture the second failure mode we are trying to measure.

Stage	Model	Configuration and objective
1. Degradation	None (deterministic operator)	k-space truncation at factor 4 followed by Rician noise at $\sigma = 0.03$. Applied on the fly, so no degraded copies are stored.
2. Baseline	The degraded input itself, and bicubic upsampling	Reference point for confirming that super-resolution helps segmentation at all.
3a. Super-resolution (distortion-optimal)	2D U-Net, 4 levels, base width 32, dropout 0.2, residual output, 1.93M parameters	Pixel L1 loss. Adam, learning rate 10^{-3} .
3b. Super-resolution (tumor-aware)	Same architecture and parameter count as 3a	Lesion-weighted L1 with weight 20 to 40, plus an optional segmentation-consistency term against the frozen segmenter.
4. Downstream segmentation	2D U-Net, base width 32, no dropout, no residual, one logit channel, 1.93M parameters	Binary cross-entropy on clean high-resolution slices, then frozen so it cannot adapt to either super-resolution model.
5. Uncertainty	Reuse of 3a or 3b	Dropout left active at inference over 10 Monte Carlo passes. The per-pixel standard deviation across passes is the uncertainty map.

The segmenter is deliberately frozen before either super-resolution model is trained. It is the measuring instrument for both safety rates, so allowing it to keep learning would let it compensate for a lesion that a super-resolution model had erased.

Contribution and evaluation

Standard super-resolution maximizes pixel fidelity, but distortion and perceptual quality are formally at odds (Blau and Michaeli, 2018), and a small tumor is a tiny fraction of the image, so erasing it barely changes PSNR. A model can therefore score well while removing what matters, an erasure and hallucination risk documented for deep medical image reconstruction and translation (Antun et al., 2020; Cohen et al., 2018). We test this and offer a fix.

The comparison. Train the two models and **hold image quality equal** (match PSNR and SSIM), then compare what happens to the tumor. Because quality is matched, any difference in the safety metrics is due to the objective, not to how sharp the images look.

Safety metrics, stratified by tumor size (both flagged by Johnson and Lui, 2023):

- **False Negative Erasure Rate (pseudo-normalization):** how often real tumor, especially the small enhancing region, is blurred away.
- **False Positive Detection Rate (hallucination):** how often tumor-like structure is fabricated outside the true mask.

The expected, defensible finding: the quality-optimal model erases small tumors that the tumor-aware model preserves, so PSNR is not a safety metric. As a baseline check we confirm super-resolution improves segmentation over the low-resolution input (Zhou et al., 2022). We also test whether high uncertainty coincides with the residual errors, and benchmark CPU-only inference latency and memory for deployment.

Scope. A real low-field scan differs from a downsampled high-field image in contrast, noise, and distortion, not only resolution. We approximate the noise with the Rician model and treat this as a proof of concept on simulated degradation; validation on real low-field or BraTS-Africa data is the next step, not a claim we make here.

Preliminary results (synthetic proof of concept)

We trained both super-resolution U-Nets and the segmentation U-Net on simulated low-field data and evaluated on a held-out synthetic set ($n = 64$, disjoint from the training seed). These numbers are a proof of concept on simulated degradation, not a clinical result.

Safety on the held-out set. The tumor-aware objective roughly halves tumor erasure, and medium and large lesion erasure drops to zero.

Metric	Distortion-optimal	Tumor-aware
Tumor erasure rate	27.8%	15.2%
Small-lesion erasure rate	88.2%	70.6%
Dice	0.576	0.481
PSNR (dB)	26.23	19.89
False positive rate	0.343	0.615
Uncertainty AUROC	0.898	0.926

Two caveats we state plainly. In this raw comparison the tumor-aware PSNR is 6.3 dB lower, so it is not a matched-quality row; the matched-quality claim is the curve below. The false positive rate nearly doubles (0.34 to 0.62), so the tumor-aware model trades some hallucination for less erasure, a tradeoff a clinician should weigh.

Quality-safety curve (matched quality). Sweeping degradation severity and comparing at similar image quality, the tumor-aware model erases fewer lesions at every level, most clearly at moderate degradation.

Degradation	Distortion erasure	Tumor-aware erasure
$\times 3$	13.5%	1.9%
$\times 4$	25.0%	9.6%
$\times 5$	28.8%	23.1%

This supports the central claim: at matched image quality the standard objective erases small tumors that the tumor-aware objective preserves, so PSNR is the wrong target for safe enhancement. The Monte Carlo dropout uncertainty map is predictive of error (AUROC near 0.9), and inference runs on CPU.

Feasibility and plan

Everything uses open data and open models and runs with light training. **Data:** BraTS carries both high-resolution images to degrade and tumor masks for the downstream step; a subset is on Kaggle with no registration, and IXI adds extra high-resolution scans. **Models:** small U-Nets we train (SwinIR and EDSR are architecture references); training our own enables uncertainty maps and avoids the 3D, four-modality mismatch. **Compute:** 2D slices, a few epochs; Monte Carlo dropout is a handful of extra passes; final inference benchmarked on CPU.

Day	Goal
1	Build the forward degradation model (k-space truncation + Rician noise), bicubic baseline, and BraTS data loader.
2	Train the two super-resolution U-Nets; confirm both beat bicubic and match each other on PSNR and SSIM.
3	Train the 2D segmentation U-Net; run on low-resolution, both super-resolved, and true images; compute Dice.
4	Add Monte Carlo dropout uncertainty maps and build the comparison figure.
5	At matched quality, measure erasure and detection rates by lesion size; test uncertainty vs errors; benchmark CPU.
6–7	Write up, prepare figures, rehearse the demo, buffer.

Success criteria. At matched PSNR and SSIM, the tumor-aware model has a lower False Negative Erasure Rate than the quality-optimal model, especially for small lesions, supporting the claim that image quality is the wrong target. A clear, honest result is a success even if the effect is smaller than expected.

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